

SanTed Quantum Scheduling

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ADDENDUM: QUANTUM-ENABLED OPTIMIZATION FOR NEW MEXICO TRANSPORTATION SCHEDULING

Submitted in support of NMQTA FY26 Application | Statewide multi-modal scope

1. Problem and Public-Sector Context

New Mexico operates one of the most geographically dispersed and structurally underutilized public transportation networks in the country. Scheduling across operators is currently solved by heuristic and human-in-the-loop methods that cannot jointly optimize across modes, fare instruments, and time horizons. In-scope systems include:

- **Commuter rail:** NM Rail Runner Express (NMDOT/Rio Metro/Herzog), 15 stations, ~2,300 weekday boardings, Belen–Santa Fe.
- **State commuter bus:** NMDOT Park & Ride — 12 routes connecting Los Alamos, Española, Santa Fe, ABQ, Las Vegas, Las Cruces, WSMR, El Paso.
- **Regional Transit Districts:** Rio Metro, North Central (Blue Bus), South Central, and Southwest — each scheduling independently.
- **Federal 5311(f) intercity:** Greyhound/Flix (incl. new ABQ–Farmington–Durango), Jefferson Lines, Limousine Express, Navajo Transit interlines.
- **Municipal & tribal:** ABQ RIDE, Santa Fe Trails, Las Cruces RoadRUNNER, Atomic City Transit, Red Apple Transit, Navajo Transit.

Vehicle assignment, crew rostering, transfer synchronization, fleet repositioning, and demand-responsive flex routing across these operators is NP-hard at statewide scale — the same problem class (constrained QUBO / mixed-integer programming) on which near-term quantum hardware demonstrates measurable advantage today.

2. Quantum and Quantum-Enabling Computational Approach

SQS reformulates each operator's scheduling problem as a constrained QUBO / Ising model and submits it to D-Wave's Leap quantum annealing service — actual quantum computation on operational hardware. Gate-model and hybrid solvers are evaluated as complements where problem structure favors them.

Approach	Hardware / Vendor	Best-fit NM scheduling problem
Quantum annealing (lead method)	D-Wave Advantage / Advantage2 via Leap	Statewide vehicle-block assignment, Park & Ride driver rostering across 12 routes, multi-corridor fleet repositioning.
Hybrid classical–quantum	D-Wave Leap Hybrid; quantum-inspired tensor networks	Multi-operator decomposition: Benders / Dantzig-Wolfe split into classical master + quantum sub-solvers per region.
Gate-model / QAOA	IBM Quantum, IonQ Forte, Quantinuum H-series	Smaller, high-constraint subproblems: Rail Runner ↔ Park & Ride transfer synchronization at NM 599, South Capitol, ABQ Sunport.

3. Measurable Outcomes and New Mexico Economic Impact

- **Operational & fiscal:** 5–12% reduction in vehicle-hours scheduled and 8–15% improvement in transfer reliability vs. current baselines. At ~\$3.00/vehicle-mile and ~\$225K/bus-year fully loaded, a 7% efficiency gain across the Park & Ride fleet alone yields seven-figure annual recurring savings to the State.
- **Ecosystem alignment:** Direct collaboration pathways with LANL, Sandia (QSCOUT), UNM CQuIC, and the NM Quantum Moonshot — anchoring quantum-applications jobs in-state.
- **Workforce & presence:** NM headquarters, NM-resident quantum engineers hired, NMDOT pilot beginning Santa Fe ↔ Los Alamos ↔ ABQ.

4. Pilot Scope and 12-Month Milestones

Four phases against a defined initial corridor — Santa Fe ↔ Los Alamos ↔ Albuquerque — chosen because it spans NMDOT Park & Ride, Rail Runner, and NCRD operations and contains the highest-value commuter flow in the state (LANL workforce). All work completed within twelve months; final report within thirteen months.

Phase	Months	Activity	Milestone / Deliverable
1. Scope & Data	M1–M3	Corridor selection with NMDOT/Rio Metro; data pipeline from GTFS feeds, AVL, and operator scheduling exports; baseline metrics established.	Signed MOUs; baseline KPI report.
2. QUBO Build	M4–M6	Formulate vehicle-block, driver-rostering, and transfer-synchronization problems as QUBOs; validate on D-Wave Leap; classical baseline benchmark.	Working quantum pipeline; benchmark report.
3. Parallel Pilot	M7–M9	Run quantum-derived schedules in parallel with production schedules on the Santa Fe ↔ Los Alamos ↔ ABQ corridor; collect deltas across four KPIs.	Side-by-side performance data.
4. Scale Plan	M10–M12	Synthesize findings; develop multi-corridor scaling roadmap; transition partners (NMDOT, RTDs) to commercial engagement.	Final report; signed scaling agreement.

5. Team, Technical Foundation, and Partnerships

- **Founder & Principal:** Vernon T. Cox — operational transportation experience plus direct technical work on the SQS quantum optimization stack.
- **Working stack:** Functional classical scheduling MVP plus quantum pipeline integrated with D-Wave Leap and Ocean SDK; gate-model evaluation via IBM Quantum and IonQ cloud access.
- **NM ecosystem partnerships:** Engagement pathways with LANL Information Sciences (D-Wave applications), Sandia QSCOUT testbed, and UNM CQuIC.
- **Capital backing:** Institutional CDFI capital secured — satisfying NMQTA's previously-secured-funding requirement; the award is not first money in.

6. Budget Allocation

Category	Amount	% of Award	NMQTA cap compliance
Quantum compute (D-Wave Leap, IBM, IonQ)	\$58,000	29%	No applicable cap.
Software development & integration (contract)	\$54,000	27%	No applicable cap.
NM-resident payroll (key technical hire)	\$50,000	25%	At 25% payroll cap.
Data infrastructure, security, hosting	\$20,000	10%	No applicable cap.
Travel & ecosystem events	\$10,000	5%	Under 10% cap.
Legal, compliance, NM business setup	\$8,000	4%	No applicable cap.
TOTAL	\$200,000	100%	All caps respected.

7. Risks, Mitigations, and NMQTA FY26 Alignment

- **Quantum hardware / advantage risk:** Hybrid solver fallback (D-Wave Leap Hybrid + classical CP-SAT) ensures the pipeline produces production schedules even during quantum-resource downtime or where quantum advantage is not measurable. The deliverable is a working benchmarked engine, not a guaranteed speedup.
- **Operator data-access risk:** GTFS public feeds and AVL exports are sufficient for Phase 1; deeper operator data integrated as MOUs land. Phase 1 deliverables are achievable from public data alone.

SQS uses quantum superposition and entanglement to perform computation on production hardware — not a speculative claim of future integration. The award is not first money in; SQS has secured prior capital and will maintain New Mexico headquarters and physical presence for the required two-year post-award period.